Course notes

The set of complex numbers

Definition · The set $\mathbb{C}$. We take for granted the existence of a set written $\mathbb{C}$, called the set of *complex numbers*, such that:

1. $\mathbb{C}$ contains $\mathbb{R}$;
2. $\mathbb{C}$ contains a number written i such that $i^2 = -1$;
3. $\mathbb{C}$ has an addition and a multiplication that extend those of $\mathbb{R}$ and follow the same rules of computation (commutativity, associativity, distributivity);
4. every complex number z can be written in a unique way as $z = a + ib$ with a and b real.

The expression $z = a + ib$ is the *algebraic form* of z. The real number a is the *real part* of z, written $\mathrm{Re}(z)$; the real number b is the *imaginary part* of z, written $\mathrm{Im}(z)$. If $b = 0$, z is real; if $a = 0$, z is *purely imaginary*.

Property · Equality of two complex numbers. Two complex numbers are equal if and only if they have the same real part and the same imaginary part. In particular, $a + ib = 0$ with a, b real is equivalent to $a = b = 0$.

Watch out. The imaginary part of $3 - 5i$ is -5, not $-5i$: it is a real number. There is no order compatible with the operations on $\mathbb{C}$: we never write $z < z'$ for non-real complex numbers.

Example 1 · Computations in algebraic form. $(2 + 3i) + (1 - i) = 3 + 2i$ and $(2 + 3i)(1 - i) = 2 - 2i + 3i - 3i^2 = 5 + i$.

The powers of i repeat every four: $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. Hence $i^{4k+r} = i^r$ for every integer $k \geq 0$.

Conjugate

Definition · Conjugate. The *conjugate* of $z = a + ib$ (a, b real) is $\overline{z} = a - ib$.

Property · First properties. For every $z = a + ib$:

$$z + \overline{z} = 2\,\mathrm{Re}(z), \qquad z - \overline{z} = 2i\,\mathrm{Im}(z), \qquad z\overline{z} = a^2 + b^2.$$

Hence z is real if and only if $\overline{z} = z$, and purely imaginary if and only if $\overline{z} = -z$. The product $z\overline{z}$ is a nonnegative real number, zero only for $z = 0$.

Proof. $z\overline{z} = (a + ib)(a - ib) = a^2 - (ib)^2 = a^2 + b^2$. The first two equalities are obtained by adding or subtracting $a + ib$ and $a - ib$. □

Property · Inverse of a nonzero complex number. Every nonzero complex number $z = a + ib$ has an inverse:

$$\frac{1}{z} = \frac{\overline{z}}{z\overline{z}} = \frac{a - ib}{a^2 + b^2}.$$

Method · Writing a quotient in algebraic form. Multiply the numerator and the denominator by the conjugate of the denominator. Example: $\frac{1}{3+4i} = \frac{3-4i}{9+16} = \frac{3}{25} - \frac{4}{25}i$.

Property · Conjugate and operations. For all complex numbers z, z' and every natural number n:

$$\overline{z+z'} = \overline{z} + \overline{z'}, \qquad \overline{zz'} = \overline{z}\,\overline{z'}, \qquad \overline{z^n} = \overline{z}^n,$$

and if $z' \neq 0$: $\overline{\left(\frac{1}{z'}\right)} = \frac{1}{\overline{z'}}$ and $\overline{\left(\frac{z}{z'}\right)} = \frac{\overline{z}}{\overline{z'}}$.

Proof. *Product.* Let $z = a + ib$ and $z' = a' + ib'$. Then $zz' = (aa' - bb') + i(ab' + a'b)$, so $\overline{zz'} = (aa' - bb') - i(ab' + a'b)$. On the other hand, $\overline{z}\,\overline{z'} = (a - ib)(a' - ib') = (aa' - bb') - i(ab' + a'b)$. The two results are equal.

Inverse. If $z' \neq 0$, $z' \times \frac{1}{z'} = 1$. By the property of the product, $\overline{z'} \times \overline{\left(\frac{1}{z'}\right)} = \overline{1} = 1$, so $\overline{\left(\frac{1}{z'}\right)}$ is the inverse of $\overline{z'}$. The quotient follows by writing $\frac{z}{z'} = z \times \frac{1}{z'}$.

Power. By induction on n. For $n = 0$: $\overline{z^0} = \overline{1} = 1 = \overline{z}^0$. If $\overline{z^n} = \overline{z}^n$ for some integer n, then $\overline{z^{n+1}} = \overline{z^n \times z} = \overline{z^n} \times \overline{z} = \overline{z}^n \times \overline{z} = \overline{z}^{n+1}$. □

Equations

Method · Linear equation. If $a \neq 0$, the equation $az = b$ has a unique solution $z = \frac{b}{a}$, which we write in algebraic form. Example: $(1+i)z = 3 - i$ gives $z = \frac{(3-i)(1-i)}{2} = \frac{2-4i}{2} = 1 - 2i$.

Method · Equation involving z and $\overline{z}$. Let $z = x + iy$ with x, y real, write each side in algebraic form, then identify real parts and imaginary parts. This gives a system of two real equations.

Example 2 · Solving $2z + i\overline{z} = 3 - 3i$. With $z = x + iy$: $2x + 2iy + i(x - iy) = (2x + y) + i(x + 2y)$. Identification gives $2x + y = 3$ and $x + 2y = -3$, hence $x = 3$ and $y = -3$. The solution is $z = 3 - 3i$. Check: $2(3 - 3i) + i(3 + 3i) = 6 - 6i + 3i - 3 = 3 - 3i$.

Binomial formula

Property · Binomial formula. For all complex numbers a, b and every natural number n:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

Proof. By induction on n. For $n = 0$, both sides equal 1. Assume the formula holds for some integer n. Then

$$(a+b)^{n+1} = (a+b)\sum_{k=0}^{n}\binom{n}{k}a^k b^{n-k} = \sum_{k=0}^{n}\binom{n}{k}a^{k+1}b^{n-k} + \sum_{k=0}^{n}\binom{n}{k}a^k b^{n+1-k}.$$

In the first sum, set $j = k+1$: it becomes $\sum_{j=1}^{n+1}\binom{n}{j-1}a^j b^{n+1-j}$. Grouping the terms with the same exponent of a and using Pascal's rule $\binom{n}{j-1} + \binom{n}{j} = \binom{n+1}{j}$ for $1 \leq j \leq n$, we obtain

$$(a+b)^{n+1} = b^{n+1} + \sum_{j=1}^{n}\binom{n+1}{j}a^j b^{n+1-j} + a^{n+1} = \sum_{j=0}^{n+1}\binom{n+1}{j}a^j b^{n+1-j}.$$

The formula holds for $n+1$, hence for every natural number n. □

Example 3 · A power of $1+i$. $(1+i)^4 = 1+4i+6i^2+4i^3+i^4 = 1+4i-6-4i+1 = -4$. One can also write $(1+i)^2 = 2i$, then $(1+i)^4 = (2i)^2 = -4$.

Key point · Key points. Unique algebraic form; $i^2 = -1$; $z\overline{z} = a^2 + b^2$; for a quotient, multiply by the conjugate of the denominator; for an equation in z and $\overline{z}$, set $z = x + iy$ and identify.